Section 2.8 — Rational Functions — Answer Key

Math 103: College Algebra

Identifying Rational Expressions

Identify each expression as rational or not.

1. Yes
2. No
3. Yes ($x + \sqrt{2}$ is a polynomial!)
4. Yes

Negative Exponents

Rewrite each expression involving negative exponents as a rational expression.

5. $\dfrac{2y^2}{x^4 z}$
6. $\dfrac{2w^5}{3u^2}$
7. $\dfrac{12}{ab^7}$
8. $\dfrac{j^3}{4k^3}$

Rational Functions

For each pair of functions f and g, write the function $\left(\frac{f}{g}\right)(x)$ as a rational function, and find its domain. **Do not simplify** your rational function.

9. $\left(\frac{f}{g}\right)(x) = \dfrac{2x^2 - 5x + 3}{2x^2 - 5x - 7}$ $\quad$ Dom $\frac{f}{g} = \left\{x \mid x \neq -1, \frac{7}{2}\right\}$
10. $\left(\frac{f}{g}\right)(x) = \dfrac{2x - 3}{4x^2 - 9}$ $\quad$ Dom $\frac{f}{g} = \left\{x \mid x \neq -\frac{3}{2}, \frac{3}{2}\right\}$

Evaluating Rational Functions

Let $h(x) = \dfrac{2x^2 - 5x + 3}{x^3 - 3x^2 + 3x - 3}$. Find

11. $h(2) = -1$
12. $h(-3) = -\frac{6}{11}$
13. $h(\frac{1}{2}) = -\frac{8}{17}$
14. $h(1)$ is undefined

Let $s(x) = \dfrac{x^2 - 5x - 11}{2x^2 + 5}$

15. $s(7) = \frac{3}{103}$
16. $s(-1) = -\frac{5}{7}$
17. $s(1.5) = -\frac{65}{38}$
18. $s(0) = -\frac{11}{5}$

Simplifying Expressions

Simplify each expression. You may leave your answer in factored form.

19. $\dfrac{2k}{3h}$
20. $-\dfrac{5a}{b^3}$
21. $-\dfrac{1}{2x}$
22. $\dfrac{y}{4x^5u^4z^2}$
23. $\dfrac{3}{5}$
24. $\dfrac{(x+4)(x^2+1)}{(x-4)(x+1)^2}$
25. -1
26. $\dfrac{2}{3}$
27. $-\dfrac{1}{2x}$
28. $\dfrac{3b-4}{5b-1}$
29. $\dfrac{p+q}{m-n}$
30. $a + b$
31. $\dfrac{x+y}{x^2+xy+y^2}$
32. $\dfrac{x-4}{x+2}$
33. $\dfrac{2(2n^2-5)}{n+1}$
34. $\dfrac{x-2}{x}$

Simplifying Functions

Simplify each function. Note the domain restrictions, if any. You may leave your answer in factored form.

35. $f(x) = \dfrac{1}{x-3}$, $x \neq 0$
36. $g(x) = \dfrac{1}{4(x+2)}$, $x \neq 1$
37. $h(x) = \dfrac{1}{x+1}$, $x \neq \frac{7}{3}$
38. $k(x) = \dfrac{x+3}{2x-1}$, $x \neq 3$
39. $s(x) = \dfrac{4x-3}{x^2-2x+4}$, $x \neq -2$
40. $t(x) = \dfrac{2x+3}{9-4x}$, $x \neq -2$